NEURAL CODEC LATENT DISTILLATION FOR CORPUS RETRIEVAL SYNTHESIS ON THE ELK STOMP

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ABSTRACT

Latent granular resynthesis encodes a corpus and an input with a neural audio codec, replaces each input frame with a similar corpus frame and decodes the result. The Elk Stomp Dev Kit, without a neural processing unit, cannot run the codec used here, music2latent, so it plays stored audio grains instead. We ask whether a small query encoder, distilled from music2latent selects grains as that encoder would, and what then limits real-time operation. A distilled multilayer perceptron of 103k parameters per frame is compared offline on input with four other query encoders, and each analysis stage is timed on the ELK. It preserves the encoder's similarity ordering better than a linear map and MFCCs (Spearman ρ .83 against .80 and .58), with selections on average in the encoder's top 4–7 %. On the ELK it takes about 1 % of the 93 ms analysis period, and only the search grows with corpus size.

1. INTRODUCTION

The Elk Stomp Dev Kit runs the low-latency Elk Audio OS [1] on an STM32MP157 with two Cortex-A7 cores at 800 MHz without NPU). Latent granular resynthesis [2] encodes an input and a corpus of recordings with a neural audio codec, replaces each input frame with a similar corpus frame and decodes the result. Here the codec is music2latent [3], a consistency autoencoder which turns each ≈93 ms of audio into a latent which is then decoded into audio. With 58 M parameters and no official 32-bit armv7 LibTorch build, neither part can run on the ELK. The latent granular instrument was originally built on a M1 Max Macbook for the Music Hackspace × MUTEK Montréal hackathon and presented it at its winners' showcase. [1] The corpus can be encoded offline so only the live input, needs encoding on the ELK. We ask whether a small query encoder distilled from the music2latent encoder (the teacher) selects grains as the teacher would, and what then limits real-time operation. The student is a multilayer perceptron trained to reproduce the teacher's latents. We hypothesize that the student preserves the teacher's ordering of all grains by similarity better than a linear map and MFCCs (H1). We additionally hypothesize that on the ELK it costs a small fraction of the time between analyses while the search over all grains becomes the largest cost for large corpora (H2). Experiments 1 and 2 test H1 offline on four corpora and H2 on the ELK.

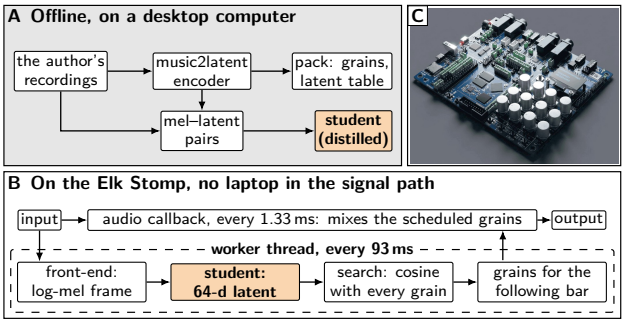


Figure 1. (A) Offline, a student network is distilled from the music2latent encoder, which encodes the author's recordings. (B) On the Elk Stomp, with no laptop in the signal path, the student replaces the encoder for the live input, and stored grains replace the decoder. (C) The Elk Stomp Dev Kit. Image: Elk Audio.

2. RELATED WORK

Neural audio instruments select grains by similarity in a codec's latent space [2], morph sounds via codec tokens [4] or navigate an autoencoder's latent space [5]. All three decode through the model and have no embedded hardware. Instruments with embedded processing [1] need no laptop during performance. Elk Audio OS lists support for ML applications, including real-time music information retrieval, among its requirements [1]. It runs models on the Raspberry Pi 4 [6, 7]. RTNeural [8] has run neural audio effects in real time on a phone with four Cortex-A7 cores [9], the Elk Stomp's core type. Knowledge distillation trains a small student to reproduce a larger teacher's outputs [10]. For embeddings the student regresses the frozen teacher's vectors under a squared error [11, 12] or cosine loss [13]. The distilled audio encoders of [11–14] are evaluated by word error rate or classification, not by

[1] https://forum.mutek.org/en/shows/2026/music-hackspace-x-mutek-hackathon-winners-showcase

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whether they keep the teacher's similarity ordering, which retrieval depends on and H1 tests.

3. METHOD

The desktop instrument replaces each input latent with a grain's latent, sampled from a softmax over cosine similarities with temperature τ, so repeated phrases vary. On the ELK (Fig. 1), the student replaces the encoder for the live input and stored audio grains joined with 10 ms crossfades, replace the decoder. The ELK thus concatenates recorded audio, as concatenative synthesis does [15], and gives up the interpolation between grains that Tokui and Baker attribute to decoding [2]. A pack holds a corpus's 16-bit audio and teacher latents. [2] [3]

The student takes a frame's 80 band log-mel spectrum chosen because the ELK computes it with one 4096-point FFT and a filterbank. It has two 256-unit hidden layers and runs through RTNeural [8]. It was trained on 45,983 mel–latent pairs from the author's recordings and MUSDB18 previews under MSE $+ 0.3\,(1 -$ mean-centred cosine$)$.

Real-time operation. The audio callback, every 64 samples at 48 kHz (1.33 ms), only mixes scheduled grains and never waits for the analysis. A worker thread runs the analysis every 4,480 samples. Each analysis runs the front end (resampling, mel frame), the student, the search (cosine similarity to every grain) and grain selection. Real time here means that each callback and each analysis finishes within its period.

4. EXPERIMENTS

Experiment 1 compares five query encoders. Three are small enough for the ELK: a linear ridge map from log-mel bands and spectral descriptors, the student, and a 128-unit GRU to test temporal context. Two are references as the teacher itself sampled with a second seed and query MFCCs matched against corpus MFCCs, a hand-crafted descriptor match as in concatenative synthesis [15]. The 12 queries are held out at file level from all corpora and training, as live input would be. Each is matched against four corpora (percussive, tonal, textural, field recordings).

Metrics. The first criterion is Spearman's ρ between the encoder's and the teacher's similarity orderings of all grains. The second is the selected grain's percentile rank in the teacher's ordering (.95: more similar than 95 % of grains). Selection agreement is the share of frames where an encoder selects the teacher's grain. Hub share, the share of selections on the five most selected grains measures repetition related to hubness [16]. Both are reported but not used to rank, as sampling at $\tau = 0.05$ keeps agreement low even for the teacher's second seed. Experiment 2 timed each analysis stage on the ELK for three corpus sizes (Fig. 2) and 32- or 8-bit latent tables.

[2] Code, ELK logs and notebooks: `https://github.com/Grashopr-888/ISMIR_LBD_2026_submission`

[3] Demo video: `https://youtu.be/gLobf3o0Qg0?si=mHY-RTp-Ta_0zCg6`

Query encoder	Parameters	ρ	Pct.	Agr.	Hub
Teacher, second seed	—	—	.981	.026	.17
MFCC cosine, no map	—	.575	.835	.007	.12
Ridge map	5.7 k	.797	.933	.039	.20
GRU-128	0.09 M	.826	.945	.045	.21
Student (MLP)	0.10 M	.829	.945	.047	.22

Table 1. Parameters, Spearman ρ, percentile rank (Pct.), agreement (Agr.) and hub share (Hub) over 48 query–corpus pairs at $\tau = 0.05$.

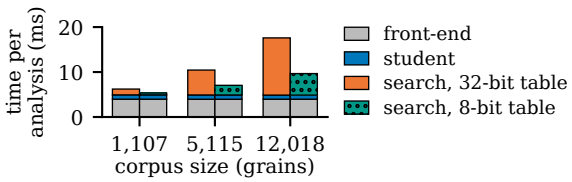


Figure 2. Median time per analysis on the Elk Stomp (period 93.3 ms). Only the search grows with the corpus, and runs 2.6 to 3.1 times faster with the 8-bit table.

5. RESULTS

Retrieval fidelity (H1). The student and the GRU keep the teacher's ordering equally well (Table 1, ρ .829 and .826). The student leads the ridge map by .032, but in every query and MFCCs lag behind (.575). Its selections rank near the teacher's (.945 against .981) showing H1 is supported.

On device cost (H2). Every analysis finished within its period and no run lost input. The callback is not the limit as granular synthesis averaged 1.95–2.39 % of it (at most 10.9 %), against 0.093 % on an M1 Max laptop. The student took about 1 % of the period while the search grew with the corpus (Fig. 2), costing most from 5,115 grains on (32-bit table). The 8-bit table kept the student's ordering ($\rho = 1.000$). H2 is supported as for large corpora the search limits how short the period can be.

6. DISCUSSION

A student of 0.10M parameters keeps most of the teacher's 58 M parameter ordering and lets the matching run on the ELK, limited by the search. The study is limited by four corpora of the author's recordings and one ELK and the metrics measure agreement with the teacher's ordering. Onsets are the student's weakest frames (cosine distance .295 against .262 for sustained frames). Future work should add an ordering term to the per frame loss, such as relational or similarity preserving losses from computer vision [17,18] or top-K ranking distillation [19]. Onsets may need stacked mel frames or an onset weighted loss.

7. AI USAGE STATEMENT

Anthropic Claude aided the initial literature review and drafting.

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